7.6.3 Relative Extrema on an Interval

A function $f(x)$ is said to have a relative maximum at x_0, if there is an open interval containing x_0 on which $f(x_0)$ is the largest value. Similarly, $f(x)$ is said to have a relative minimum at x_0, if there is an open interval containing x_0 on which $f(x_0)$ is the smallest value.

A relative maximum need not be the largest value on the entire domain, while a relative minimum need not be the smallest value on the entire domain. Therefore, there may be more than one relative maximum or relative minimum on the entire domain.

A relative extrema of a function is the extreme values (maximum or minimum) of the functions among all the evaluated values of $f(x), \forall x \in I \subset D$ where I may be open or closed. Usually the local extreme value of a function is attained at a critical point. Note that, a function may have a critical point at $x = c$ without having a local extreme value there. For instance, both of the functions $y = x^3$ and $y = x^{\frac{1}{3}}$ have critical points at the origin, but neither function has a local extreme value at the origin.

Theorem 7.9 (Fermat)

If $f(x)$ has a relative extrema at $x = c$ then c is a critical number. Invariably there will be critical numbers of the function obtained as solutions of the equation $f'(x) = 0$ or as values of x at which $f'(x)$ does not exist.

7.6.4 Extrema using First Derivative Test

After we have determined the intervals on which a function is increasing or decreasing, it is not difficult to locate the relative extrema of the function. The location or points at which the relative extrema occurs for a given function $f(x)$ can be observed through the graph $y = f(x)$. However to find the exact point and the value of the extrema of functions we need to use certain test on functions. One such test is the first derivative test, which is stated in the following theorem.

Theorem 7.10 (First Derivative Test)

Let $(c, f(c))$ be a critical point of function $f(x)$ that is continuous on an open interval I containing c. If $f(x)$ is differentiable on the interval, except possibly at c, then $f(c)$ can be classified as follows:
(when moving across the interval I from left to right)

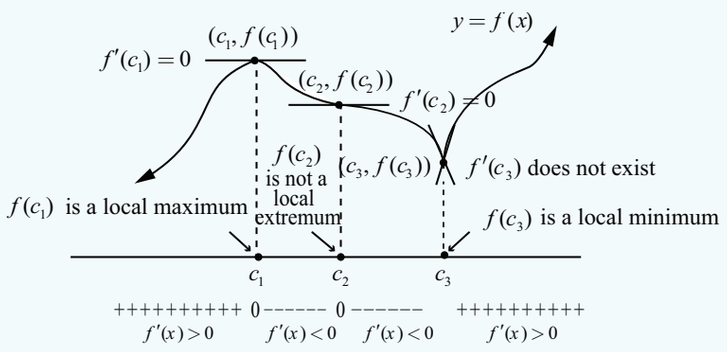


Fig. 7.20

(i) If $f'(x)$ changes from negative to positive at c, then $f(x)$ has a local minimum $f(c)$.

(ii) If $f'(x)$ changes from positive to negative at c, then $f(x)$ has a local maximum $f(c)$.

(iii) If $f'(x)$ is positive on both sides of c or negative on both sides of c, then $f(c)$ is neither a local minimum nor a local maximum.

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